AI IN HEALTHCARE: CHALLENGES & BREAKTHROUGHS

Data Science News of the Week

Chloe Barker

WHY THIS MATTERS

- Rising demand for efficient and accurate care
- Great advancements in AI and data science tools
- Significant potential to improve patient outcomes and reduce healthcare costs

QUICK OVERVIEW OF AI & DS IN HEALTHCARE

- Diagnostic imaging & computer vision
  - Detecting tumors in medical scans
  - Analysis of medical images
- Remote patient monitoring
  - Wearables
  - Real-time alerts & monitoring
- Predictive analytics
  - Identification of high-risk patients
  - Personalized treatment
  - Detection of drug interactions
- Administrative automation
  - Scheduling
  - Chatbots
  - Coding of medical notes

WHAT DOES AI IN MEDICINE LOOK LIKE?

DRUG AND DOSAGE RECOMMENDATIONS

TELEMEDICINE & REMOTE CONSULTATIONS

ANALYSIS OF VITAL SIGNS AND TEST RESULTS FOR PERSONALIZED RECOMMENDATIONS

MORE ACCURATE RISK ASSESSMENTS

DIAGNOSES OF DISEASES IN MEDICAL IMAGES (MRIS, CT SCANS, & X-RAYS)

PERSONALIZED CARE PLANS

EARLY IDENTIFICATION OF CANCER AND CHRONIC DISEASE

REAL-TIME VITALS MONITORING

AUTOMATED APPOINTMENT SCHEDULING

CHAT-GPT 4.0 VS. MEDICAL STUDENTS

Table. Summary of Cases, Clinical Skills Assessed, and Scores for Chatbot and Student Responses by Case and Clinical Reasoning Skill

		Assessed clinical reasoning skills						Chatbot score, %[a]		
Case description	Total word count	Diagnostic schema[b]	Differential diagnosis	Illness scripts[c]	Case summary	Problem list	Other[d]	Model 3.5	Model 4	Student score, mean (SD), %
Chronic fatigue and anemia	583	Yes	Yes	No	No	Yes	Yes	62.9	76.8	75.9 (10.7)
Acute abdominal pain and diarrhea	798	Yes	Yes	No	No	Yes	Yes	72.7	90.6	83.1 (11.5)
Acute confusion and hypertension	1013	Yes	Yes	Yes	Yes	Yes	Yes	65.4	88.7	78.2 (11.6)
Chronic diarrhea and amenorrhea	1109	No	Yes	No	Yes	No	Yes	65.	70.2	82.1 (11.0)
Subacute fever and abdominal pain	919	Yes	Yes	No	No	Yes	Yes	67.9	87.8	83.0 (7.7)
Chronic dyspnea	831	No	Yes	No	No	Yes	No	60.4	85.7	74.1 (14.0)
Acute chest pain	747	No	No	No	No	No	Yes	82.3	91.7	96.0 (13.0)
Acute RUQ pain	902	No	Yes	No	Yes	No	Yes	61.5	81.6	77.6 (16.7)
Acute lightheadedness	885	No	Yes	No	Yes	Yes	Yes	41.5	70.3	80.5 (12.2)
Acute abdominal pain and fever	1071	No	Yes	No	Yes	No	Yes	79.4	99.4	86.8 (10.3)
Chronic fatigue	917	No	Yes	No	Yes	Yes	Yes	71.4	94.7	86.2 (8.3)
Subacute confusion	953	Yes	Yes	Yes	Yes	Yes	Yes	71.6	93.6	78.2 (8.3)
Acute abdominal pain and nausea	972	Yes	Yes	No	No	Yes	Yes	81.1	86.5	84.1 (10.0)
Subacute dyspnea	1093	No	Yes	Yes	No	Yes	Yes	80.9	92.4	80.5 (11.2)
Model 4 score, mean (SD), %[e]	NA	89.8 (9.0)	84.3 (13.8)	92.0 (7.8)	81.9 (25.3)	87.7 (11.2)[f]	86.4 (17)	NA	NA	NA
Student score, mean (SD), %[e]	NA	85.4 (17.3)	86.1 (16.8)	87.6 (14.8)	82.2 (21.0)	71.8 (20.1)[f]	82.8 (20.3)	NA	NA	NA

Abbreviations: NA, not applicable; RUQ, right upper quadrant.

[a] Scores listed are the mean score for each case from 2 runs, each graded by 2 independent faculty graders using the same grading rubric.

[b] A diagnostic schema is defined as a thorough collection of causes for a specific symptom, which is organized into categories based on organ system or physiological process.

[c] An illness script is defined as a summary of the features of a specific disease, organized into categories such as epidemiology, historical features, examination findings, and relevant test abnormalities.

[d] Other assessed clinical skills include the following: diagnostic test selection and interpretation, identification of cognitive biases, discussion of relevant literature search strategies, and interpretation of the significance of physical examination findings.

[e] Scores listed are the mean score for questions or prompts that tested a specific clinical reasoning skill, graded by 2 independent faculty graders using the same grading rubric.

[f] The difference in mean scores between model 4 and students on problem list–related questions was statistically significant (*P* < .001).

(Strong et al., 2023)

CHAT-GPT 4.0 VS. MEDICAL STUDENTS

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Findings:

- Model 4 of the chatbot outperformed first- and second-year students from Stanford University School of Medicine on clinical reasoning examinations and had significant improvement over Model 3.5.

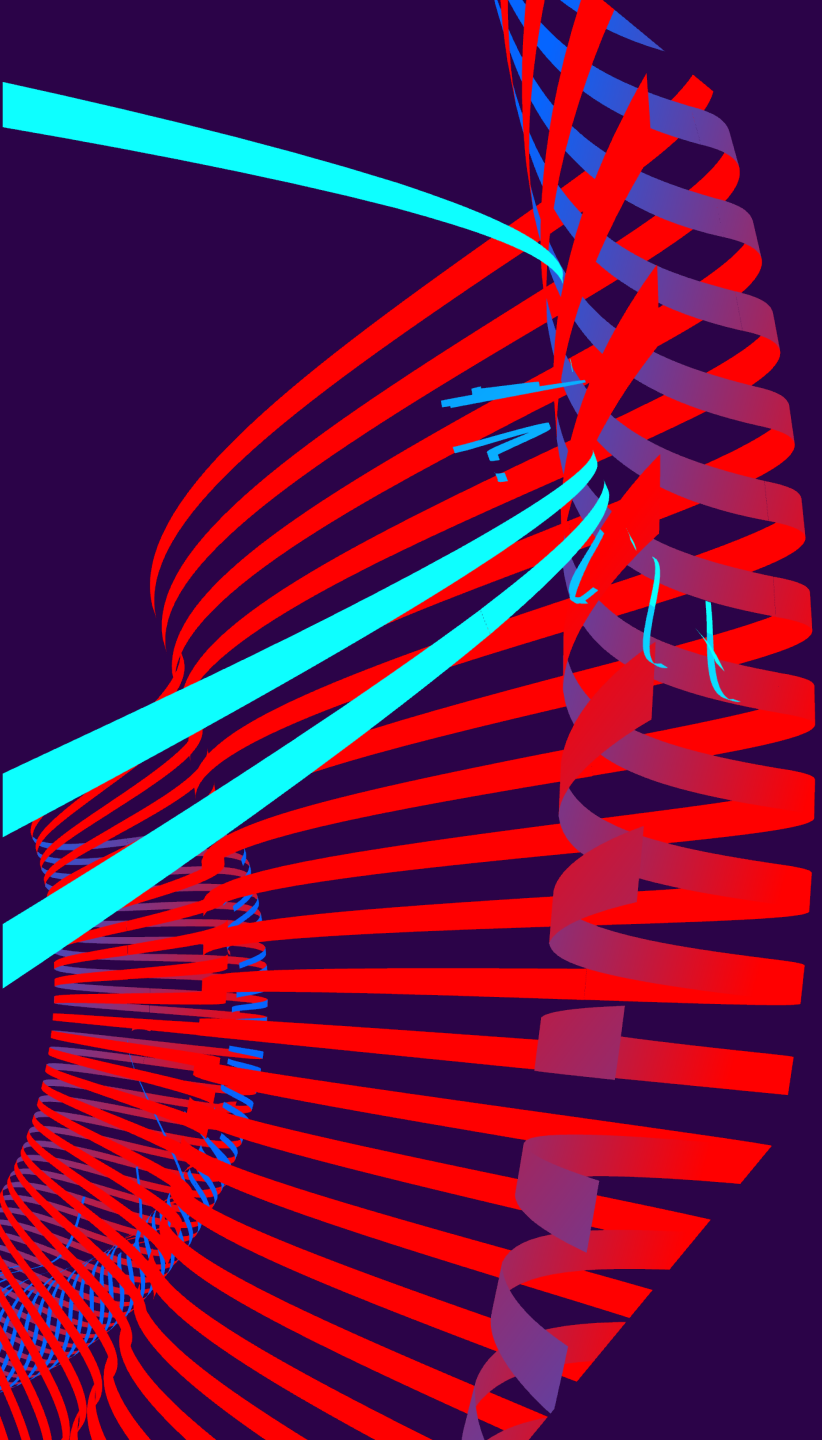
Implications:

- This suggests great potential for growth of chatbot technology over time and within the medical field.
- Medicine should integrate AI-related concepts into its curriculum to enhance medical practices and patient outcomes.

(Strong et al., 2023)

POTENTIAL BENEFITS

- Improve patient outcomes and reduce medical costs
- Increase efficiency, reproducibility, and coverage of screening
- Overcome human limitation and error
- Increase access to healthcare resources
- Provide earlier detection and other preventative measures for disease



SOME CURRENT RESEARCH

BREAST CANCER PREDICTION USING MACHINE LEARNING

- Breast cancer is the second most diagnosed cancer and the leading cause of death among in women.
- Strong limitations in breast cancer diagnosis: mammography has around 70% accuracy on average and there is biopsy-related human error.
- A recently published study used eight machine learning classifiers with 11 features that may contribute to breast cancer prognosis on a dataset of 213 breast cancer patients, exploring how feature selection may affect performance in predicting breast cancer diagnoses (Almustafa & La Mogolia, 2025).

Data Collection and Description: The dataset of observations was obtained from the University of Calabar Teaching Hospital cancer registry over 24 months (January 2019 - August 2021).

(Almustafa & La Moglia, 2025)

BREAST CANCER PREDICTION USING MACHINE LEARNING

Metric of performance (original dataset and third feature selection).

Classifier	Original Dataset		Third Feature Selection	
	Accuracy (Train)	Accuracy (Test)	Accuracy (Train)	Accuracy (Test)
Logistic Regression	90.14	91.94	88.44	89.06
Random Forest	100	90.32	99.32	89.06
Extra Trees	100	88.71	99.32	89.06
XGB	100	87.10	96.60	89.06
LGMB	98.59	87.10	94.56	92.19
Cat Boost	99.30	90.32	93.88	90.63
SVC	91.55	88.71	79.59	79.69
Gaussian Naïve Bayes	91.55	88.71	89.12	90.63

Third Feature Selection : repeat each classifier with their four most important features in predicting the output.

Highest performing accuracies (original data):

- Logistic Regression: 91.94 %
- Random Forest classifier: 90.32 %

Highest performing accuracy (third feature selection):

- LGMB: 92.19 %

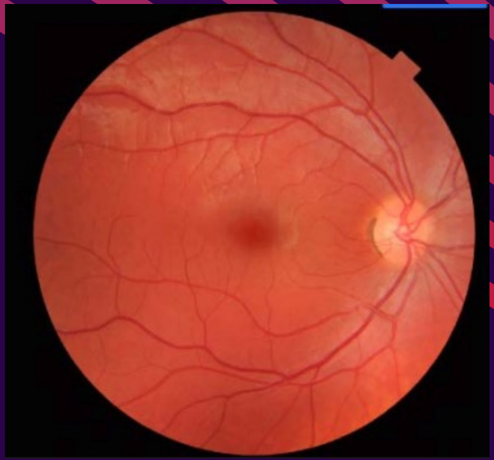
(Almustafa & La Moglia, 2025)

IMPLICATIONS

- Machine learning can enhance or surpass traditional diagnostic methods, significantly improving accuracy and reducing errors caused by human limitations.
- It supports healthcare professionals by providing faster, data-driven insights, resulting in quicker and more accurate breast cancer diagnoses.
- The potential of machine learning extends beyond breast cancer detection, offering improved predictive capabilities for various diseases, such as heart disease and diabetes, thereby transforming patient care and outcomes.

(Almustafa & La Moglia, 2025)

DEEP LEARNING ALGORITHM TO DETECT DIABETIC RETINOPATHY

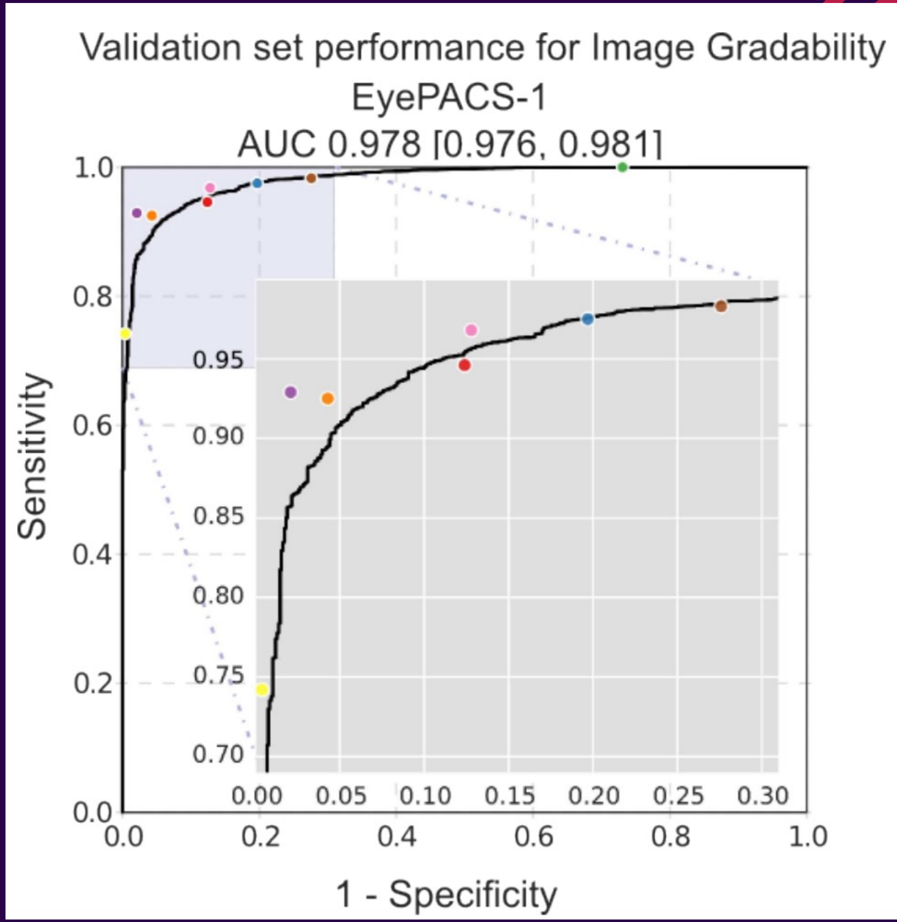


Deep learning : a predictive algorithm that uses datasets of images to develop patterns and output predictions.

The EyePACS-1 dataset consisted of 9963 images of retinal fundus from 4997 patients with diabetes used to train an algorithm to detect referable diabetic retinopathy.

Sensitivities of 97.5% and 96.1% were achieved with these deep-learning models.

Implications: Great potential for using these models in a clinical setting, as they provide evidence that they meet the standards of the best doctors and beat the standards set by average doctors (Gulshan et al., 2016).



Performance of the Algorithm (Black Curve) and Ophthalmologists (Colored Dots) for Predicting Gradable Images on the EyePACS-1 Dataset (9946 Images)

(Gulshan et al., 2016)

RISKS AND CHALLENGES

Data privacy/security

Protection of sensitive medical data and patient records from breaches.

Bias & data quality issues

Risk of inaccurate or biased data influencing healthcare models and algorithms.

Regulatory hurdles (e.g. FDA approvals)

Challenges in meeting regulatory standards for data integration and clinical model deployment.

Clinician/patient acceptance and trust

Resistance to change and low trust in new healthcare technologies.

FUTURE OUTLOOK

Better data integration & explainable AI for trust

Larger, more diverse datasets to reduce bias and have generalizable results

Collaboration across medicine, computer science, and policy

KEY TAKEAWAYS

- AI is rapidly reshaping healthcare diagnostics, monitoring, and personalization
- Strong promise shown in recent studies (e.g., breast cancer prediction, deep-learning in diagnosis)
- Ethical, regulatory, and data challenges remain

REMEMBER!

AI models are powerful educational tools that can complement, rather than replace, medical professionals - greater integration of AI education can empower healthcare workers, improve clinical decision-making, and ultimately enhance patient care.

THANK YOU

Chloe Barker

chloebarker@smu.edu

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